



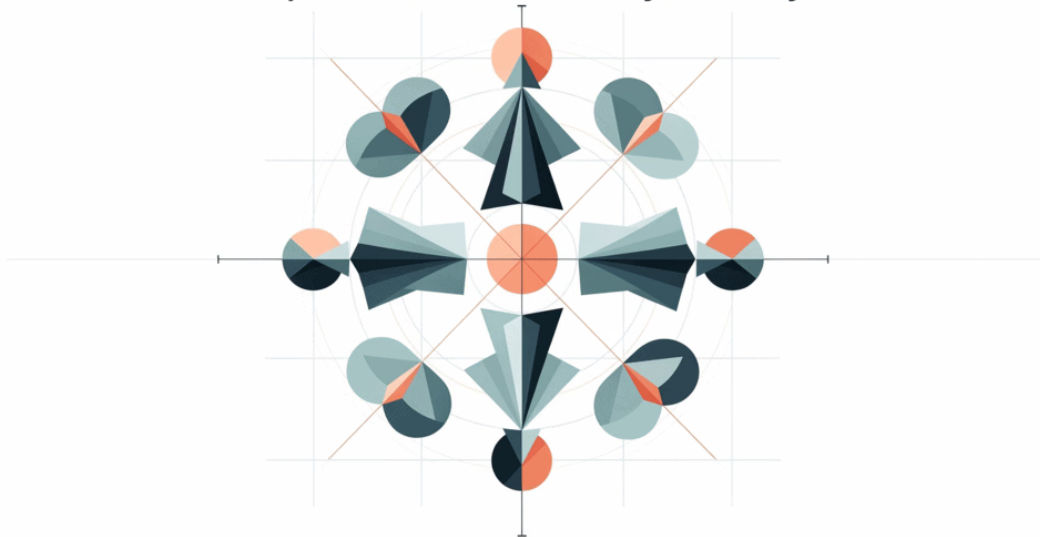
Important Formulas: Symmetry

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Important Formulas: Symmetry



Symmetry is a fundamental concept in geometry used to identify patterns and properties of shapes.

1. Lines of Symmetry (Reflectional Symmetry)

A line of symmetry divides a figure into two identical halves.

1.1 Number of Lines of Symmetry

Square $\rightarrow 4$

Rectangle $\rightarrow 2$

Circle $\rightarrow \infty$ (infinite)

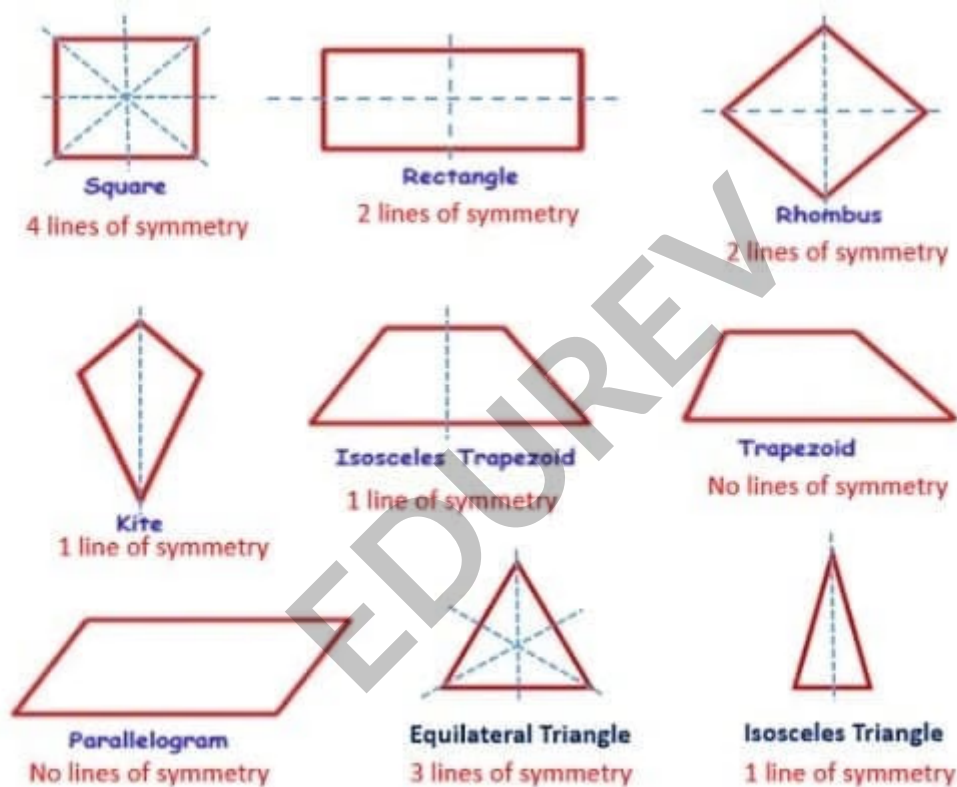
Equilateral Triangle $\rightarrow 3$

Isosceles Triangle $\rightarrow 1$

Regular Pentagon $\rightarrow 5$

Regular Hexagon $\rightarrow 6$

Regular Polygon $\rightarrow n$



1.2 Formula (Regular Polygon)

Number of lines of symmetry = n

1.3 Test for Line Symmetry

Fold along the line $\rightarrow$ halves must coincide exactly.

2. Rotational Symmetry

A shape has rotational symmetry if it looks identical after rotation about a fixed point.

2.1 Key Terms

Center of Rotation

Angle of Rotation

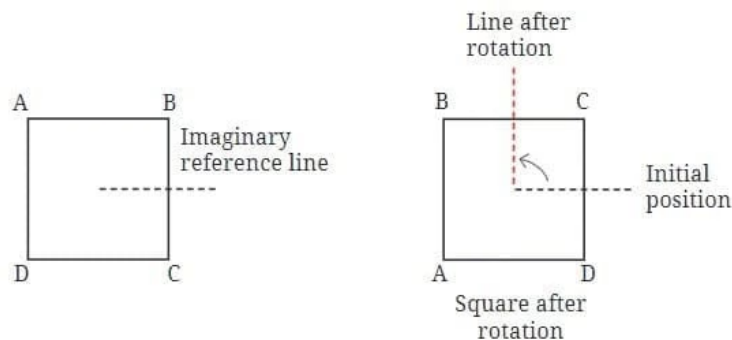
Order of Rotational Symmetry

2.2 Standard Angles

Full Turn $\rightarrow 360^\circ$

Half Turn $\rightarrow 180^\circ$

Quarter Turn $\rightarrow 90^\circ$



2.3 Formula for Angle of Rotation

$$\theta = 360^\circ \div n$$

Where:

θ = angle of rotation

n = order of rotational symmetry

2.4 Orders for Common Shapes

Square $\rightarrow 4$ ($90^\circ, 180^\circ, 270^\circ, 360^\circ$)

Rectangle $\rightarrow 2$ ($180^\circ, 360^\circ$)

Equilateral Triangle $\rightarrow 3$ ($120^\circ, 240^\circ, 360^\circ$)

Circle $\rightarrow \infty$

Pentagon $\rightarrow 5$ ($72^\circ, 144^\circ, 216^\circ, 288^\circ, 360^\circ$)

Hexagon $\rightarrow 6$ ($60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ, 360^\circ$)

2.5 Formula (Regular Polygon)

Order of rotational symmetry = n

3. Radial Symmetry (Fans / Windmills)

3.1 Angle Between Arms

Angle between arms = $360^\circ \div n$

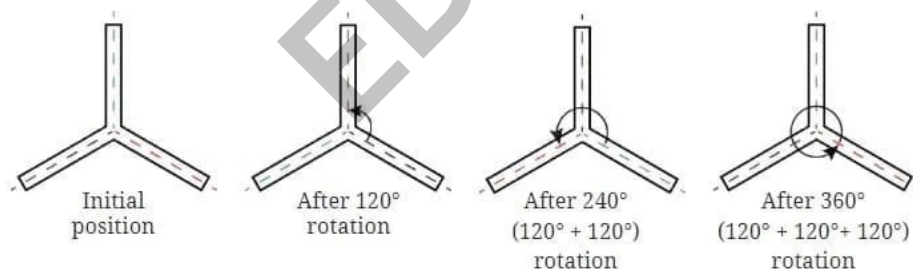
Example:

4 arms $\rightarrow 90^\circ$

3 arms $\rightarrow 120^\circ$

3.2 Angles of Symmetry

Angles = $(360^\circ \div n)$, $2 \times (360^\circ \div n)$, $3 \times (360^\circ \div n)$, ..., 360°

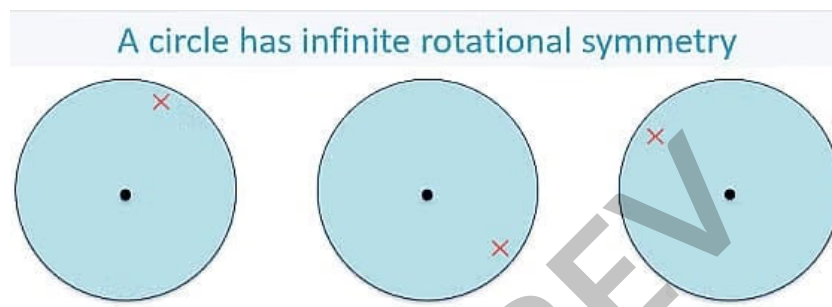


4. Special Case: Circle

Lines of symmetry $\rightarrow \infty$

Order of rotational symmetry $\rightarrow \infty$

Angle of rotation $\rightarrow$ Any angle 0° to 360°



5. Quick Formulas

5.1 Line Symmetry

Lines of symmetry (regular polygon) = n

5.2 Rotational Symmetry

$$\theta = 360^\circ \div n$$

$$\text{Order} = n$$

5.3 Radial Figures

$$\text{Angle between arms} = 360^\circ \div n$$

6. Common Mistakes

Rectangle has 2, not 4 lines

Symmetry only when halves exactly match

Formula valid only for regular polygons

Every shape has symmetry at 360°

Rotational symmetry $\neq$ always line symmetry

7. Summary

Regular polygon $\rightarrow$ lines = order = n

Circle $\rightarrow$ most symmetric figure

Use formulas to quickly solve problems

